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CHE 425

Midterm Review Paper

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Nanotherapeutics Administered by Inhalation:

Abstract

When treating cancer, first thing that comes to mind is chemo radiation and oral medication. But when treating lung cancer, inhalation drug delivery has been found to be the most effective method is targeting the lung cancer cell tumors. This review first focuses on the differences between inhalation drug delivery method versus orally administered chemotherapeutic drugs. It then goes into the pulmonary pharmacokinetic processes. This explains the six crucial steps of how inhalation is processed and absorbed through the body. It also explains the effectiveness in targeting tumor cells in the lung. Although this drug delivery comes with many challenges such as not being able to make all chemotherapy drugs effectively accessible through inhalation, there are clinical trials in the works of making dry powder inhalers.

1. **Introduction:**

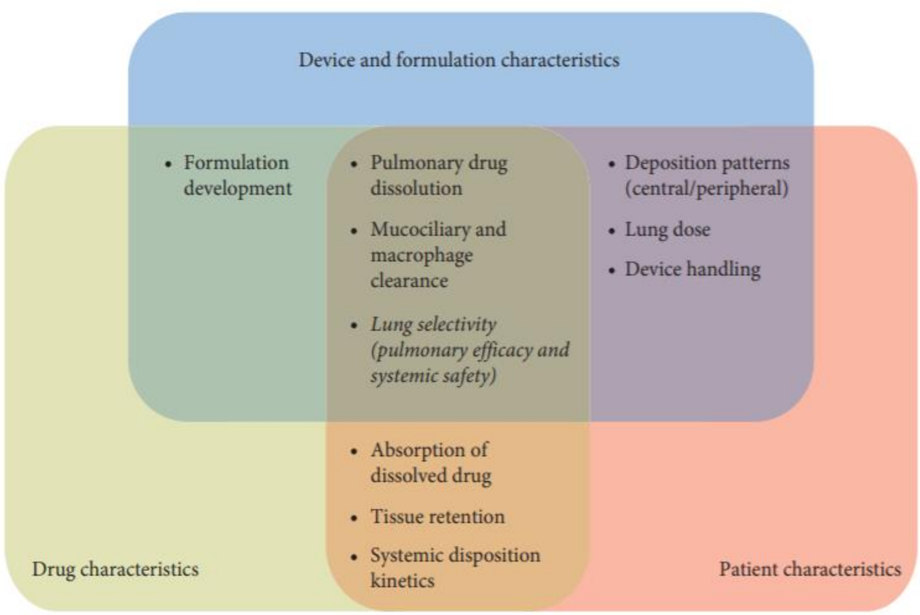
Chemotherapeutic drugs administered by inhalation have been proven to be more effective then orally administered drugs in lung cancer patients. Inhalation is the act of breathing in the medication sending it straight to the lungs rather than orally digesting it. Lung cancer is the leading cause of all cancer deaths in both genders in the United States of America. There has been much speculation on which delivery of medicine to the lungs has the best effect on this dangerous disease. The main idea behind this is that inhalation delivery goes directly to the lung tumor cells. However, oral medication will evidently first be absorbed by small intestine moving to the liver then into the blood stream, then distributed throughout the body, then metabolized and excreted.

Inhalation drugs are usually used to treat diseases such as asthma and chronic obstructive pulmonary disease (COPD), but studies have recently found that inhalation drugs can be affective in chemotherapeutic drugs. Chemotherapeutic drugs that are orally consumed are not as effective as the ones administered by inhalation because they are not always reaching the lung cancer tumor cells, whereas with inhalation, it goes directly into the lungs and targets the cancer. Inhalation drugs for respiratory diseases such as lung cancer include corticosteroids, beta-sympathomimetic, muscarinic antagonists, and antibiotics. (3) The inhalation delivery method has one primary goal which allows it to have such gained importance over the past decades. This goal is to reduce pulmonary symptom by inhibiting inflammation and constriction of lung airways. (4) A benefit to using the inhalation method to treat lung cancer, a type of pulmonary disease, is using smaller doses because the medication directly targets the disease rather than taking high doses of regular oral medication that doesn’t reach the tumor cells. (6) Using smaller doses of inhalation also decreases the risk of the medication side effects and leads to higher pulmonary efficacy. (5)

2. Pulmonary Pharmacokinetic Processes

The most important factor to distinguish between the inhalation delivery method and the oral medication method is how long it takes for the medication to have a positive effect. The faster the effect on the lungs, the better the method of delivery is. In this case, I can compare two different types of medicines. We can look at albuterol for the inhalation delivery method and Salmeterol for the oral medication delivery. Albuterol can make its impact in approximately a couple of minutes while salmeterol can take up to a half an hour (7). This tie difference can be critical, and for all we know can be the difference between life and death for the lung cancer patient. According to Dr. Chris Carter, a Senior Lecturer the Strathclyde Institute of Pharmacy and Biomedical Sciences, “By delivering cisplatin, one of the most widely used drugs for lung cancer, in a vaporized form, we would be able to get it to the cancerous cells and avoid the damage to healthy cells which can be hugely debilitating to patients.” It is believed that this treatment would be easier on the patient than going through chemo radiation and will allow them to live a longer and fuller life. This shows us how the inhalation method has gained much popularity, because it will not damage any healthy cells a patient has, where the oral medication route does give that risk. (11)

However, in order for the inhalation method to work we much understand the properties, design, and development of these drugs that will ensure the best possible outcomes for the patients. We must analyze the inhalation drug itself, the device for which the drug is being inhaled from, and lastly the characteristic of the patient on how they are inhaling this drug. The following figure will show how these three major components overlap when it comes it this delivery method.



The following figure below gives a summary of the pharmacokinetic process in the lungs for the inhaled drug. This will explain how the process flows in two different regions in the lungs. The conducting airways as the mouth/throat region where oral medication is taken through. The alveolar region is what the inhaled drug would evidently go through. Later on in this paper we will analyze each step of the process and how it correlates to each specific region.

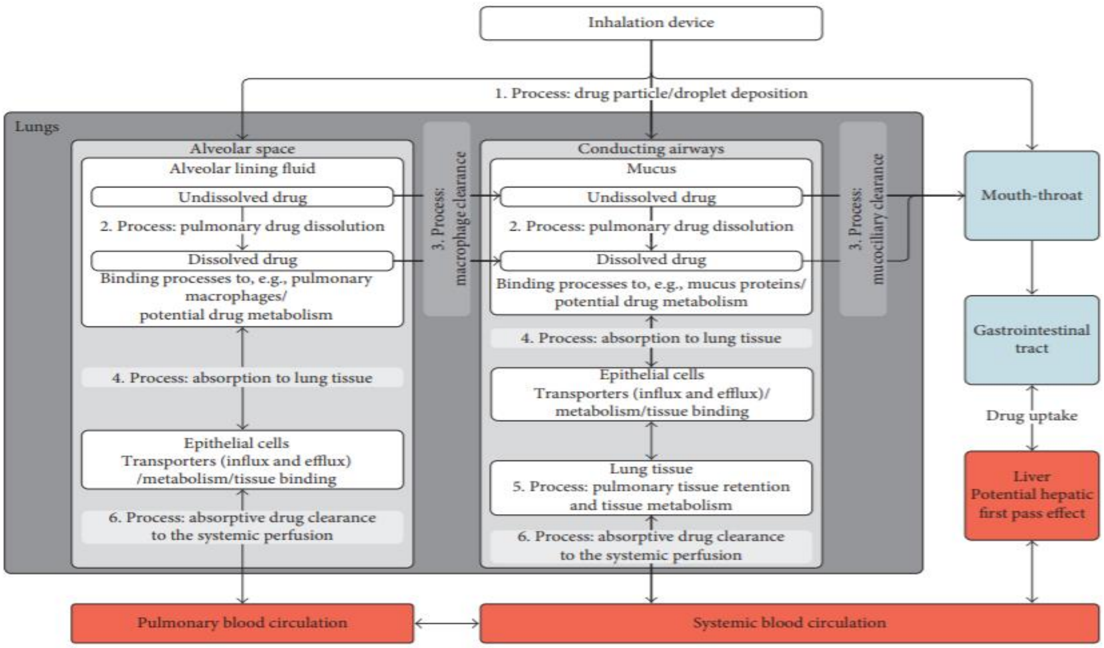
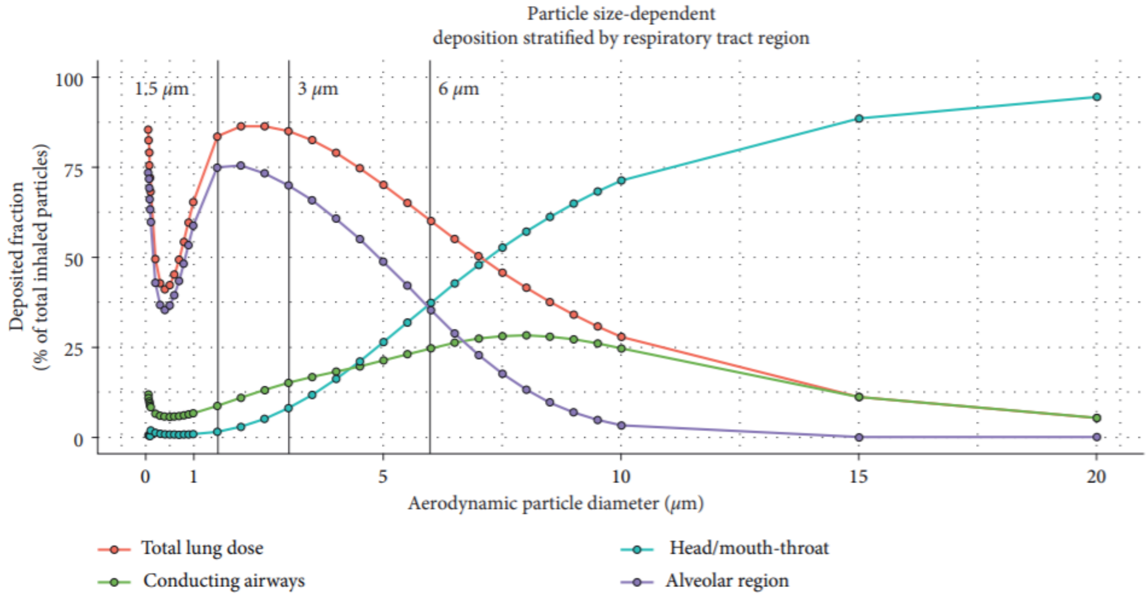


Figure 1: Summary of the lung-specific PK processes for inhaled drugs. Overview of the pulmonary-specific kinetic processes (1–6). The direction of the arrows indicates the direction of each process. For example, drug dissolution is considered to be a unidirectional process.

As seen above, the pharmacokinetic process in the lungs from the inhaled drug involves six major steps according to the figure above. The process involves, “1. drug particle or droplet deposition; 2. drug dissolution in the lung fluids; 3. mucociliary clearance in the conducting airways and macrophage clearance in the alveolar space; 4. absorption (of dissolved drug) to the lung tissue; 5. pulmonary tissue retention and potential pulmonary metabolism; and 6. absorptive drug clearance (drug transport) from the lung tissue to the systemic perfusion” (3).

The first step is the drug particle or droplet deposition which occurs right after the patient has inhaled the drug from the device. This is will depend on the inhalation drug itself, the device for which the drug is being inhaled from, and lastly the characteristic of the patient on how they are inhaling this drug. At this instance, one fraction of the dose inhaled will remain in the device while the other fraction will go to the respiratory system. The drug particles will go through the airways in the lungs this will eventually lead to the deposition of the drug. This will normally take place in the airway itself, mouth, and/or throat. The drug dose which gets deposited in the lung and the pulmonary patterns is dependent on four major things. These include the particle size of the drug, inhalation flow, and type of device being, and the factor of the pulmonary disease.



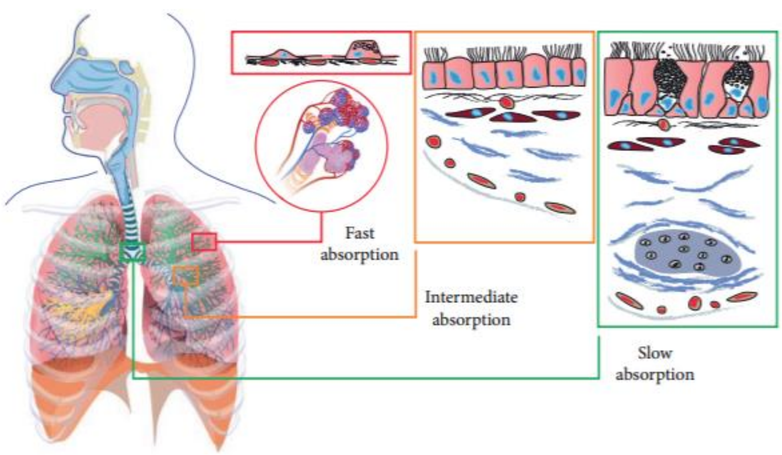
As we see in the figure above, the size of the drug particle will generally tell you where the particle will be deposited once it enters the respiratory system. A particle size from the range of 0.5um-5um will have a greater chance of being deposited in the lung and giving a high lung dose, which is critical. However, particle size greater than 5um will most likely end up in the mouth or throat. (8) This will decrease the lung does and may bring down the impact of the drug itself because it is not reaching the spot it needs to be at.

This is where we analyze the inhalation device itself. The goal is to now design specific devices which help discharge particle size of drugs that are in the appropriate range of 1um-5um, taking them to the lung site rather than the throat/mouth site. This will eventually maximize the lung and lead to better impact/effect of the drug itself. Lastly, we analyze the way this patient is inhaling this drug. If the patient inhales the drug at a fast rate, this could lead to the deposit of the drug in the mouth or throat region. In order for the drug to reach the lung site, a slower rate of inflation is preferred. Therefore, a device that combines a slow inhalation of the drug with a small size of the drug particle will evidently give the best method of inhalation delivery. (9)

The second step is where will analyze the pulmonary drug dissolution. This is the process of how the drug itself will dissolve in the fluids of the epithelial lining once it has reach the target site, which is the lung. This is all dependent on three major components. These are the drug formulation, physicochemical drug properties, and the physiologic factors. (10) The airways for which the drug particles lie in, is lined with biphasic gel-aqueous mucus layer which will evidently be a barrier to the drug particle dissolution because of its thickness (12). However, there are alveolar cells which produce surfactant, these will make the tension on the surface smooth enough to where the dissolution of the drug particle will be made easier (13). This explains how the drug formulation can affect the dissolution process. Next component is the properties of the drug itself. The desired dissolution can be slow or fast, depending on what makes the lung retention take longer because it will evidently help the patient. Therefore, when it comes to drug properties, the dissolution profile being fast or slow is necessary in determining the type of drug needed.

The third step is the mucociliary clearance in the conducting airways and macrophage clearance in the alveolar space. In order to insure a high bioavailability of the deposited drug particles which will lead to a high lung dose, there must be an effective clearance of any harmful particles in the lung. This clearance is the mucociliary clearance, which is a mechanism that will get rid of any bacteria or dust particles that might affect the lung dosage. This defense mechanism is at its fastest with large airways and mucus layers. So any drug particles in the central airways in the lungs get removed, while drug particles in any other parts of airways will get moved up to the central airways where they can actually be absorbed. However, this can be different depending on the characteristics of the patient. The drugs formulation, characteristics, and the characteristics of the patient all effect this defense mechanism. The other clearance is the macrophage; this is not used because it's a very slow defense mechanism for inhaled drugs to be removed.

The fourth step is the absorption of the drug particle to the lung tissue. The drug particles that make it through the clearance described above and dissolve into the lining fluid will eventually get absorbed in the tissue of the lung. This step is dependent on two major components. These components are the patient's characteristics of the airway and the characteristics of the drug being used. Drugs which are lipophilic are known to have rapid absorption, however drugs which are hydrophilic it is not as fast (14-15). Airways in patients can have different surface areas, perfusion, and epithelial thickness (16-19). In the next figure, it can be see how specific airways regions affect absorption of drug particles.



The two areas which will be compared is the alveolar region and the regular conducting airways such as mouth or throat. Since the alveolar space region has a greater surface area and perfusion, which leads it to have a higher absorption rate of the drug particles. (16,14) The epithelial thickness in the alveolar region is thinner than the mouth/throat region. The thinner the layer is the higher absorption rate as show in the figure above.

The fifth step is the pulmonary tissue retention and potential pulmonary metabolism. When the drug particle is absorbed into the tissue, it is heavily influenced by the drugs characteristics and the airways characteristic of the patient. (21) Inhaled drugs such as long acting muscarinic receptor antagonists (LAMAs) or long acting β2-agonists (LABAs) are known to have low permeability and will generally have lung retention that is very long. (21) When it comes to inhaled drugs, the desired result is to have a long pulmonary retention time. This will lead to a longer time for the pulmonary efficacy (24). Due to this long retention time, drug particles have the ability to be redistributed to other airway regions which is another reason of a higher pulmonary efficacy is occurring. This will evidently benefit the patient very heavily, and show the importance of having a longer tissue retention.

The sixth step which is the last step is the absorptive drug clearance (drug transport) from the lung tissue to the systemic perfusion. This is where the adoptive drug clearance of the drug particle to go from the lung tissue to the blood circulation. However, in order for this clearance process to be most effective it need to occur in an airway region of heavy perfusion. The region in the lung with the highest local perfusion is the alveolar region. As described before, this region has a very high surface area and a think epithelium, which makes it a high absorption are and will evidently lead to the absorptive drug clearance process occurring at a faster rate than this occurring in the conducting airways such as mouth/throat. (16) This will evidently conclude the pulmonary pharmacokinetic process for an inhaled drug.

3. Conclusion:

In conclusion, lung cancer is the deadliest type of cancer in the world. The way to treat this disease can be more effective, staring with delivery method of the medicine itself. We see how the process occurs for an inhaled drug and why it generally leads to high pulmonary efficacy and decreasing the risk of side effects better than oral medication can do. This occurs in pharmacokinetic process in the lungs from and involves the six major explained above. The first step was the drug particle or droplet deposition where a device that combines a slow inhalation of the drug with a small size of the drug particle will evidently give the best method of inhalation delivery. The second step was the drug dissolution in the lung fluids where we see how the drug formulation can affect the dissolution process. The third step was the mucociliary clearance in the conducting airway and how it is influenced by the drugs formulation, characteristics, and the characteristics of the patient and can evidently lead to a higher bioavailability of the drug particle and a more effective lung dose. The fourth step was the absorption (of dissolved drug) to the lung tissue where a greater surface area and perfusion, and thinner epithelium specifically in the alveolar region leads to a higher absorption rate. The fifth step was the pulmonary tissue retention and potential pulmonary metabolism where we see how it is affected by drug characteristics of inhaled drugs such as having low permeability which evidently leads to the desired result of a long pulmonary retention time. This will lead to a longer time for the pulmonary efficacy. Lastly, the sixth step was the absorptive drug clearance (drug transport) from the lung tissue to the systemic perfusion where we see how the alveolar region is where the absorptive drug clearance process occurs at a faster rate than this occurring in the conducting airways such as mouth/throat. Although not all chemotherapy drugs are in inhalation form, there are clinical trials on dry powder inhalers (DPIs) that have been said to be promising chemotherapy drugs because they deliver high drug doses within a short time period of less than a minute. (24)

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